Object Oriented Programming (CST-004) - Unit 5

Unit 5 - File I/O, Namespaces & STL - notes in easy English, matched to the UTU syllabus.

Study Materials by Anas - study-materials-by-anas.vercel.app

What this unit covers: file input/output, namespaces, and a first look at the STL.

Key ideas: File I/O uses streams: **ifstream** (read), **ofstream** (write), **fstream** (both). Open with a mode - ios::in, ios::out, ios::app (append), ios::binary - check is_open(), use << / >> or get()/put(), getline() for whole lines, then close(). **Namespaces** group names to avoid clashes: namespace std is why we write std::cout or add using namespace std. **STL** (Standard Template Library) gives ready containers - **vector** (dynamic array), **list** (doubly linked list), **map** (key → value, sorted), **set** (unique sorted values) - plus **iterators** to walk them and **algorithms** like sort() and find().

Exam pointers: a program that writes data to a file and reads it back is the standard question - practise it with getline. Short theory: file opening modes table, "what is a namespace and why", and "name three STL containers with one use each".

Made by Anas. Free forever. Watch the linked videos on the site after reading.